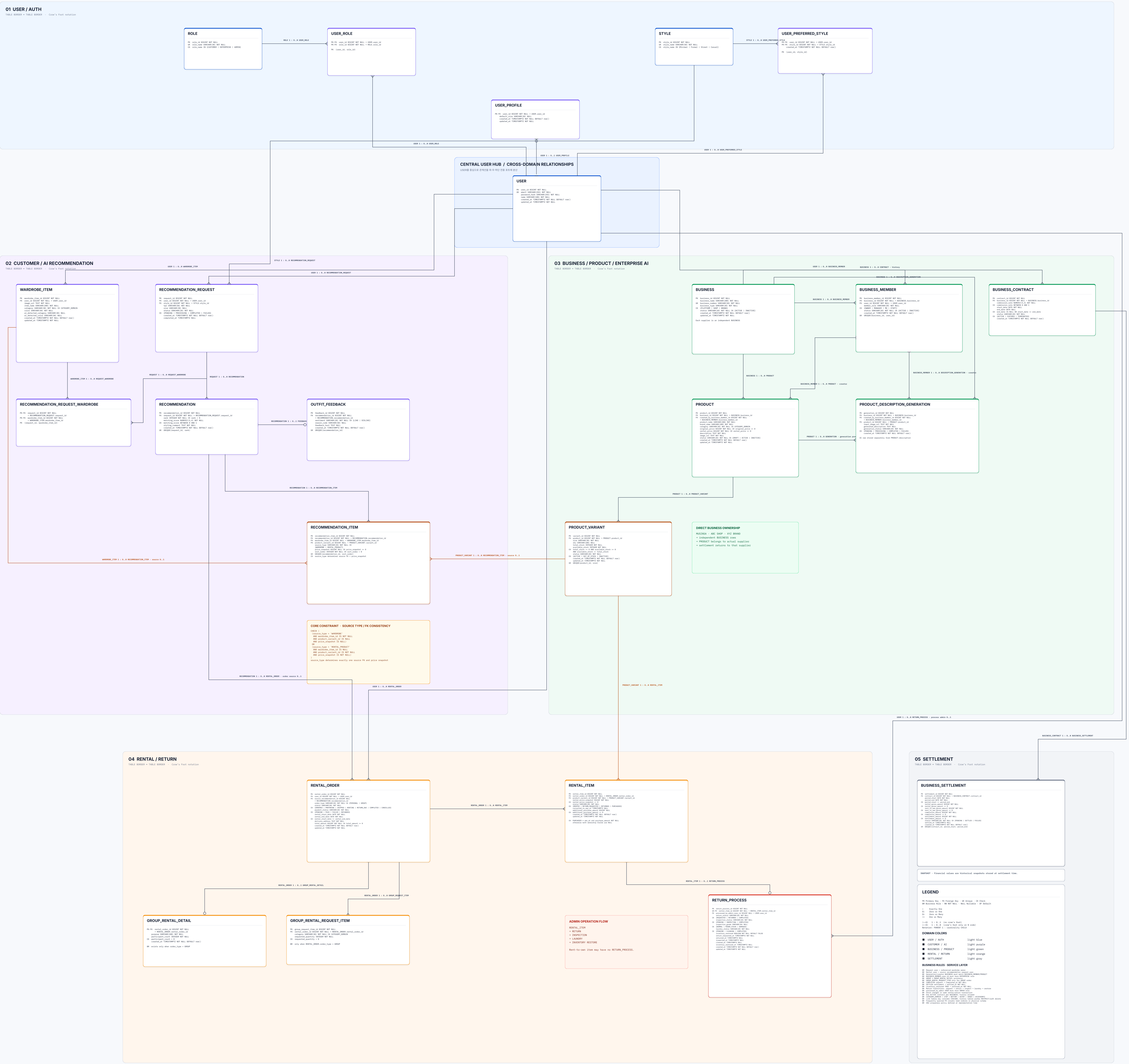
FITLY · LOGICAL ENTITY RELATIONSHIP DIAGRAM

Implementation-ready PostgreSQL model · Data types · NULL / DEFAULT · CK / UK · Service-layer BR


ROLE 1 : 0..N USER_ROLE
STYLE 1 : 0..N USER_PREFERRED_STYLE
USER 1 : 0..N USER_ROLE
USER 1 : 0..1 USER_PROFILE
USER 1 : 0..N USER_PREFERRED_STYLE
STYLE 1 : 0..N RECOMMENDATION_REQUEST
USER 1 : 0..N WARDROBE_ITEM
USER 1 : 0..N RECOMMENDATION_REQUEST
WARDROBE_ITEM 1 : 0..N REQUEST_WARDROBE
REQUEST 1 : 0..N REQUEST_WARDROBE
REQUEST 1 : 0..N RECOMMENDATION
RECOMMENDATION 1 : 0..1 FEEDBACK
RECOMMENDATION 1 : 0..N RECOMMENDATION_ITEM
WARDROBE_ITEM 1 : 0..N RECOMMENDATION_ITEM · source 0..1
PRODUCT_VARIANT 1 : 0..N RECOMMENDATION_ITEM · source 0..1
RECOMMENDATION 1 : 0..N RENTAL_ORDER · order source 0..1
USER 1 : 0..N RENTAL_ORDER
USER 1 : 0..N BUSINESS_MEMBER
BUSINESS 1 : 0..N CONTRACT · history
BUSINESS 1 : 0..N DESCRIPTION_GENERATION
BUSINESS 1 : 0..N BUSINESS_MEMBER
BUSINESS 1 : 0..N PRODUCT
BUSINESS_MEMBER 1 : 0..N PRODUCT · creator
BUSINESS_MEMBER 1 : 0..N DESCRIPTION_GENERATION · creator
PRODUCT 1 : 0..N GENERATION · generation proc
PRODUCT 1 : 0..N PRODUCT_VARIANT
PRODUCT_VARIANT 1 : 0..N RENTAL_ITEM
USER 1 : 0..N RETURN_PROCESS · process admin 0..1
BUSINESS_CONTRACT 1 : 0..N BUSINESS_SETTLEMENT
RENTAL_ORDER 1 : 0..N RENTAL_ITEM
RENTAL_ORDER 1 : 0..1 GROUP_RENTAL_DETAIL
RENTAL_ORDER 1 : 0..N GROUP_REQUEST_ITEM
RENTAL_ITEM 1 : 0..1 RETURN_PROCESS

01 USER / AUTH

TABLE BORDER ⇔ TABLE BORDER · Crow's Foot notation

ROLE

```
PK  role_id BIGINT NOT NULL
UK  role_name VARCHAR(30) NOT NULL
CK  role_name IN {CUSTOMER | ENTERPRISE | ADMIN}
```

USER_ROLE

```
PK·FK  user_id BIGINT NOT NULL → USER.user_id
PK·FK  role_id BIGINT NOT NULL → ROLE.role_id

PK  (user_id, role_id)
```

STYLE

```
PK  style_id BIGINT NOT NULL
UK  style_name VARCHAR(50) NOT NULL
CK  style_name IN {Minimal | Formal | Street | Casual}
```

USER_PREFERRED_STYLE

```
PK·FK  user_id BIGINT NOT NULL → USER.user_id
PK·FK  style_id BIGINT NOT NULL → STYLE.style_id
    created_at TIMESTAMPTZ NOT NULL DEFAULT now()

PK  (user_id, style_id)
```

USER_PROFILE

```
PK·FK  user_id BIGINT NOT NULL → USER.user_id
    default_size VARCHAR(30) NULL
    created_at TIMESTAMPTZ NOT NULL DEFAULT now()
    updated_at TIMESTAMPTZ NOT NULL
```

CENTRAL USER HUB / CROSS-DOMAIN RELATIONSHIPS

USER를 중심으로 관계선을 좌·우·하단 전용 포트에 분산

USER

```
PK  user_id BIGINT NOT NULL
UK  email VARCHAR(255) NOT NULL
    password_hash VARCHAR(255) NOT NULL
    name VARCHAR(100) NOT NULL
    created_at TIMESTAMPTZ NOT NULL DEFAULT now()
    updated_at TIMESTAMPTZ NOT NULL
```

02 CUSTOMER / AI RECOMMENDATION

TABLE BORDER ⇔ TABLE BORDER · Crow's Foot notation

WARDROBE_ITEM

```
PK  wardrobe_item_id BIGINT NOT NULL
FK  user_id BIGINT NOT NULL → USER.user_id
    image_url TEXT NOT NULL
    item_name VARCHAR(200) NOT NULL
    category VARCHAR(30) NOT NULL CK CATEGORY_DOMAIN
    color VARCHAR(50) NOT NULL
    ai_detected_category VARCHAR(30) NULL
    ai_detected_color VARCHAR(50) NULL
    created_at TIMESTAMPTZ NOT NULL DEFAULT now()
    updated_at TIMESTAMPTZ NOT NULL
```

RECOMMENDATION_REQUEST

```
PK  request_id BIGINT NOT NULL
FK  user_id BIGINT NOT NULL → USER.user_id
FK  style_id BIGINT NOT NULL → STYLE.style_id
    tpo VARCHAR(50) NOT NULL
    size VARCHAR(30) NULL
    status VARCHAR(20) NOT NULL
CK  {PENDING | PROCESSING | COMPLETED | FAILED}
    created_at TIMESTAMPTZ NOT NULL DEFAULT now()
    completed_at TIMESTAMPTZ NULL
```

RECOMMENDATION_REQUEST_WARDROBE

```
PK·FK  request_id BIGINT NOT NULL
       → RECOMMENDATION_REQUEST.request_id
PK·FK  wardrobe_item_id BIGINT NOT NULL
       → WARDROBE_ITEM.wardrobe_item_id
PK  (request_id, wardrobe_item_id)
```

RECOMMENDATION

```
PK  recommendation_id BIGINT NOT NULL
FK  request_id BIGINT NOT NULL → RECOMMENDATION_REQUEST.request_id
    rank INTEGER NOT NULL CK rank > 0
    matching_score NUMERIC(5,4) NOT NULL
CK  matching_score BETWEEN 0 AND 1
    styling_comment TEXT NOT NULL
    created_at TIMESTAMPTZ NOT NULL DEFAULT now()
UK  UNIQUE(request_id, rank)
```

OUTFIT_FEEDBACK

```
PK  feedback_id BIGINT NOT NULL
FK  recommendation_id BIGINT NOT NULL
    → RECOMMENDATION.recommendation_id
    sentiment VARCHAR(20) NOT NULL CK {LIKE | DISLIKE}
    reason_code VARCHAR(50) NULL
    feedback_text TEXT NULL
    created_at TIMESTAMPTZ NOT NULL DEFAULT now()
UK  UNIQUE(recommendation_id)
```

RECOMMENDATION_ITEM

```
PK  recommendation_item_id BIGINT NOT NULL
FK  recommendation_id BIGINT NOT NULL → RECOMMENDATION.recommendation_id
FK  wardrobe_item_id BIGINT NULL → WARDROBE_ITEM.wardrobe_item_id
FK  product_variant_id BIGINT NULL → PRODUCT_VARIANT.variant_id
    source_type VARCHAR(20) NOT NULL CK
    {WARDROBE | RENTAL_PRODUCT}
    price_snapshot BIGINT NULL CK price_snapshot >= 0
    sort_order INTEGER NOT NULL CK sort_order > 0
UK  UNIQUE(recommendation_id, sort_order)
CK  source_type determines source FK + price_snapshot
```

CORE CONSTRAINT · SOURCE TYPE / FK CONSISTENCY

```
CHECK (
 (source_type = 'WARDROBE'
  AND wardrobe_item_id IS NOT NULL
  AND product_variant_id IS NULL
  AND price_snapshot IS NULL)
 OR
 (source_type = 'RENTAL_PRODUCT'
  AND wardrobe_item_id IS NULL
  AND product_variant_id IS NOT NULL
  AND price_snapshot IS NOT NULL)
)
source_type determines exactly one source FK and price snapshot
```

03 BUSINESS / PRODUCT / ENTERPRISE AI

TABLE BORDER ⇔ TABLE BORDER · Crow's Foot notation

BUSINESS

```
PK  business_id BIGINT NOT NULL
    business_name VARCHAR(200) NOT NULL
UK  business_number VARCHAR(50) NOT NULL
    business_type VARCHAR(20) NOT NULL
CK  {PLATFORM | SHOP | BRAND}
    status VARCHAR(20) NOT NULL CK {ACTIVE | INACTIVE}
    created_at TIMESTAMPTZ NOT NULL DEFAULT now()
    updated_at TIMESTAMPTZ NOT NULL

Each supplier is an independent BUSINESS
```

BUSINESS_MEMBER

```
PK  business_member_id BIGINT NOT NULL
FK  business_id BIGINT NOT NULL → BUSINESS.business_id
FK  user_id BIGINT NOT NULL → USER.user_id
    member_role VARCHAR(20) NOT NULL
CK  {OWNER | MANAGER | MD | STAFF}
    status VARCHAR(20) NOT NULL CK {ACTIVE | INACTIVE}
    created_at TIMESTAMPTZ NOT NULL DEFAULT now()
UK  UNIQUE(business_id, user_id)
```

BUSINESS_CONTRACT

```
PK  contract_id BIGINT NOT NULL
FK  business_id BIGINT NOT NULL → BUSINESS.business_id
    commission_rate NUMERIC(5,4) NOT NULL
CK  commission_rate BETWEEN 0 AND 1
    start_date DATE NOT NULL
    end_date DATE NULL
CK  end_date IS NULL OR start_date <= end_date
    status VARCHAR(20) NOT NULL
CK  {ACTIVE | EXPIRED | TERMINATED}
    created_at TIMESTAMPTZ NOT NULL DEFAULT now()
```

PRODUCT

```
PK  product_id BIGINT NOT NULL
FK  business_id BIGINT NOT NULL → BUSINESS.business_id
FK  created_by_business_member_id BIGINT NOT NULL
    → BUSINESS_MEMBER.business_member_id
    product_name VARCHAR(200) NOT NULL
    brand_name VARCHAR(100) NOT NULL
    category VARCHAR(30) NOT NULL CK CATEGORY_DOMAIN
    original_price BIGINT NOT NULL CK original_price >= 0
    rental_price BIGINT NOT NULL CK rental_price >= 0
    description TEXT NOT NULL
    image_url TEXT NOT NULL
    status VARCHAR(20) NOT NULL CK {DRAFT | ACTIVE | INACTIVE}
    created_at TIMESTAMPTZ NOT NULL DEFAULT now()
    updated_at TIMESTAMPTZ NOT NULL
```

PRODUCT_DESCRIPTION_GENERATION

```
PK  generation_id BIGINT NOT NULL
FK  business_id BIGINT NOT NULL → BUSINESS.business_id
FK  created_by_business_member_id BIGINT NOT NULL
    → BUSINESS_MEMBER.business_member_id
FK  product_id BIGINT NULL → PRODUCT.product_id
    input_image_url TEXT NOT NULL
    generated_description TEXT NULL
    generation_status VARCHAR(20) NOT NULL
CK  {PENDING | PROCESSING | COMPLETED | FAILED}
    created_at TIMESTAMPTZ NOT NULL DEFAULT now()

AI raw stored separately from PRODUCT.description
```

PRODUCT_VARIANT

```
PK  variant_id BIGINT NOT NULL
FK  product_id BIGINT NOT NULL → PRODUCT.product_id
    size VARCHAR(30) NOT NULL
    sku VARCHAR(100) NULL
    total_stock INTEGER NOT NULL
    available_stock INTEGER NOT NULL
CK  total_stock >= 0 AND available_stock >= 0
    AND available_stock <= total_stock
    status VARCHAR(20) NOT NULL
CK  {ACTIVE | OUT_OF_STOCK | INACTIVE}
    created_at TIMESTAMPTZ NOT NULL DEFAULT now()
    updated_at TIMESTAMPTZ NOT NULL
UK  UNIQUE(product_id, size)
```

DIRECT BUSINESS OWNERSHIP

MUSINSA · ABC SHOP · XYZ BRAND
→ independent BUSINESS rows
→ PRODUCT belongs to actual supplier
→ settlement returns to that supplier

04 RENTAL / RETURN

TABLE BORDER ⇔ TABLE BORDER · Crow's Foot notation

RENTAL_ORDER

```
PK  rental_order_id BIGINT NOT NULL
FK  user_id BIGINT NOT NULL → USER.user_id
FK  source_recommendation_id BIGINT NULL
    → RECOMMENDATION.recommendation_id
    order_type VARCHAR(20) NOT NULL CK {PERSONAL | GROUP}
    status VARCHAR(20) NOT NULL
CK  {ORDERED | PREPARING | SHIPPED | RENTING | RETURN_DUE | COMPLETED | CANCELLED}
    payment_status VARCHAR(20) NOT NULL
CK  {PENDING | PAID | FAILED | REFUNDED}
    rental_start_date DATE NOT NULL
    rental_end_date DATE NOT NULL
CK  rental_start_date <= rental_end_date
    delivery_address TEXT NOT NULL
    total_amount BIGINT NOT NULL CK total_amount >= 0
    created_at TIMESTAMPTZ NOT NULL DEFAULT now()
    updated_at TIMESTAMPTZ NOT NULL
```

RENTAL_ITEM

```
PK  rental_item_id BIGINT NOT NULL
FK  rental_order_id BIGINT NOT NULL → RENTAL_ORDER.rental_order_id
FK  product_variant_id BIGINT NOT NULL → PRODUCT_VARIANT.variant_id
    rental_price_snapshot BIGINT NOT NULL
CK  rental_price_snapshot >= 0
    status VARCHAR(20) NOT NULL
CK  {RENTED | RETURN_REQUESTED | RETURNED | PURCHASED}
    converted_to_own_at TIMESTAMPTZ NULL
    additional_purchase_amount BIGINT NULL
CK  additional_purchase_amount >= 0
    created_at TIMESTAMPTZ NOT NULL DEFAULT now()
    updated_at TIMESTAMPTZ NOT NULL

CK  PURCHASED ⇔ own_at and purchase_amount NOT NULL
    otherwise both ownership fields are NULL
```

GROUP_RENTAL_DETAIL

```
PK·FK  rental_order_id BIGINT NOT NULL
       → RENTAL_ORDER.rental_order_id
    purpose VARCHAR(100) NOT NULL
    participant_count INTEGER NOT NULL
CK  participant_count > 0
    created_at TIMESTAMPTZ NOT NULL DEFAULT now()

BR  exists only when order_type = GROUP
```

GROUP_RENTAL_REQUEST_ITEM

```
PK  group_request_item_id BIGINT NOT NULL
FK  rental_order_id BIGINT NOT NULL → RENTAL_ORDER.rental_order_id
    category VARCHAR(30) NOT NULL CK CATEGORY_DOMAIN
    requested_quantity INTEGER NOT NULL
CK  requested_quantity > 0

BR  only when RENTAL_ORDER.order_type = GROUP
```

ADMIN OPERATION FLOW

RENTAL_ITEM
→ RETURN
→ INSPECTION
→ LAUNDRY
→ INVENTORY RESTORE

Rent-to-own item may have no RETURN_PROCESS.

RETURN_PROCESS

```
PK  return_process_id BIGINT NOT NULL
UK·FK  rental_item_id BIGINT NOT NULL → RENTAL_ITEM.rental_item_id
FK  processed_by_admin_user_id BIGINT NULL → USER.user_id
    return_status VARCHAR(20) NOT NULL
CK  {REQUESTED | RETURNED | COMPLETED}
    inspection_status VARCHAR(20) NOT NULL
CK  {PENDING | INSPECTING | COMPLETED}
    inspection_grade VARCHAR(20) NULL
CK  {NORMAL | MINOR_STAIN | DAMAGED}
    laundry_status VARCHAR(20) NOT NULL
CK  {PENDING | CLEANING | COMPLETED}
    inventory_restored BOOLEAN NOT NULL DEFAULT FALSE
    return_requested_at TIMESTAMPTZ NOT NULL
    returned_at TIMESTAMPTZ NULL
    inspected_at TIMESTAMPTZ NULL
    cleaned_at TIMESTAMPTZ NULL
    inventory_restored_at TIMESTAMPTZ NULL
    created_at TIMESTAMPTZ NOT NULL DEFAULT now()
    updated_at TIMESTAMPTZ NOT NULL
```

05 SETTLEMENT

TABLE BORDER ⇔ TABLE BORDER · Crow's Foot notation

BUSINESS_SETTLEMENT

```
PK  settlement_id BIGINT NOT NULL
FK  contract_id BIGINT NOT NULL → BUSINESS_CONTRACT.contract_id
    period_start DATE NOT NULL
    period_end DATE NOT NULL
CK  period_start <= period_end
    rental_gross_amount BIGINT NOT NULL
CK  rental_gross_amount >= 0
    rent_to_own_gross_amount BIGINT NOT NULL
CK  rent_to_own_gross_amount >= 0
    commission_amount BIGINT NOT NULL
CK  commission_amount >= 0
    settlement_amount BIGINT NOT NULL
CK  settlement_amount >= 0
    status VARCHAR(20) NOT NULL CK {PENDING | SETTLED | FAILED}
    settled_at TIMESTAMPTZ NULL
    created_at TIMESTAMPTZ NOT NULL DEFAULT now()
UK  UNIQUE(contract_id, period_start, period_end)
```

SNAPSHOT · Financial values are historical snapshots stored at settlement time.

LEGEND

PK Primary Key · FK Foreign Key · UK Unique · CK Check
BR Business Rule · NN NOT NULL · NULL Nullable · DF Default

```
|     Exactly One
O|    Zero or One
O<    Zero or Many
|<    One or Many

|——O|   1 : 0..1  (no crow's foot)
|——O<   1 : 0..N  (crow's foot only on N side)
Notation: PARENT 1 : cardinality CHILD
```

DOMAIN COLORS

■	USER / AUTH	light blue
■	CUSTOMER / AI	light purple
■	BUSINESS / PRODUCT	light green
■	RENTAL / RETURN	light orange
■	SETTLEMENT	light gray

BUSINESS RULES · SERVICE LAYER

```
BR  Request user = referenced wardrobe owner
BR  Rental user = source recommendation request user
BR  Generation/creator BUSINESS must match BUSINESS_MEMBER/PRODUCT
BR  BUSINESS_MEMBER.user_id must have ENTERPRISE role
BR  GROUP ⇔ GROUP_RENTAL_DETAIL existence
BR  GROUP_RENTAL_REQUEST_ITEM only for GROUP order
BR  COMPLETED request → completed_at NOT NULL
BR  SETTLED settlement → settled_at NOT NULL
BR  inventory_restored TRUE → restored_at NOT NULL
BR  Return transitions: request → return → inspect → laundry → restore
BR  processed_by_admin USER must have ADMIN role
BR  Stock changes in same rental/return transaction
BR  One ACTIVE contract per BUSINESS; history allowed
BR  CATEGORY_DOMAIN = {TOP | BOTTOM | OUTER | SHOES | ACCESSORY}
BR  Link tables may consider CASCADE; history tables prefer RESTRICT/soft delete
BR  Frequently queried FK columns need indexes in physical schema
BR  SKU uniqueness policy defined at implementation time
```

FITLY ERD · 24 confirmed tables · PostgreSQL types / NN / NULL / DF / CK / UK / BR · No new entities